

INFORMATION RETRIEVAL – QUESTION BANK CHECKLIST

Module 1 – Introduction to Information Retrieval

1. Define Information Retrieval and list classification of IR systems. [2] 5m
2. Explain the difference between Information and Data with examples. [3] 5m
3. Discuss the relationship of IR systems with DBMS, digital libraries and data warehouses. [1] 10m
4. Explain the retrieval process in digital libraries. [2] 5m
5. Explain trends and research issues in Information Retrieval. [1] 5m
6. Illustrate an Information Retrieval system. [1] 10m

Module 2 – Modeling in Information Retrieval

1. Draw the taxonomy of IR models and explain any one modelling technique. [3] 10m
2. Explain the taxonomy of Information Retrieval Models. [2] 10m
3. Discuss Classic vs Probabilistic IR models. [2] 10m
4. Explain Algebraic, Theoretical and Probabilistic models. [1] 10m
5. Compare Boolean Model and Vector Space Model. [2] 5m
6. List key components of the Vector Space Model. [1] 5m
7. Explain Vector Space Model for scoring with example. [2] 10m
8. Explain Latent Semantic Indexing Model with advantages. [3] 10m
9. Discuss advantages and disadvantages of Classic IR vs Alternative Models. [1] 10m
10. Explain Probabilistic Latent Semantic Indexing (PLSI). [1] 10m

Module 3 – Query and Operations in Information Retrieval

1. Explain the process of Structured Text Retrieval Model. → Times Asked: 4 | Marks: 5 / 10
2. Illustrate different types of queries in IR. → Times Asked: 4 | Marks: 5 / 10
3. Explain Pattern Matching in IR with example. → Times Asked: 3 | Marks: 5 / 10
4. Explain Query Processing in Distributed IR. → Times Asked: 2 | Marks: 10
5. Explain the need for Pattern Matching in IR. → Times Asked: 1 | Marks: 5
6. Write short note on Automatic Local vs Global Analysis. → Times Asked: 2 | Marks: 10
7. Explain User Relevance Feedback and its significance. → Times Asked: 3 | Marks: 5 / 10
8. Elaborate on Rocchio method for query expansion. → Times Asked: 1 | Marks: 10

- 9 Explain Keyword-based and Structured Queries. → Times Asked: 1 | Marks: 5
- 10 Describe Query Processing Loop with an example. → Times Asked: 1 | Marks: 10

Module 4 – Indexing and Scoring in Information Systems

- 1 Explain Inverted File Indexing with example. → Times Asked: 3 | Marks: 5 / 10
- 2 What is an Inverted File? Explain with diagram. → Times Asked: 2 | Marks: 10
- 3 Explain the role of Suffix Array and Suffix Tree in IR. → Times Asked: 3 | Marks: 10
- 4 Explain Parametric and Zone Indexes. → Times Asked: 3 | Marks: 10
- 5 Explain Weighted Zone Scoring. → Times Asked: 1 | Marks: 10
- 6 What is the significance of tf and idf? Explain with formula. → Times Asked: 3 | Marks: 10
- 7 Explain how tf-idf weighting helps in ranking. → Times Asked: 2 | Marks: 10
- 8 Can tf-idf weight exceed 1? Why? → Times Asked: 2 | Marks: 5 / 10
- 9 Explain Scoring and Ranking in Vector Space Model. → Times Asked: 1 | Marks: 10
- 10 Write note on Efficient Scoring and Inexact Top K Retrieval. → Times Asked: 2 | Marks: 10

Module 5 – Evaluation of Information Retrieval Systems

- 1 Explain Evaluation of IR Systems using Standard Test Collections. → Times Asked: 2 | Marks: 10
- 2 Compare Evaluation of Ranked vs Unranked Retrieval Sets. → Times Asked: 3 | Marks: 10
- 3 Explain Precision and Recall Metrics in Evaluation. → Times Asked: 1 | Marks: 10
- 4 Discuss System Quality and User Utility in IR. → Times Asked: 1 | Marks: 10
- 5 Explain System Issues and Refining a Deployed System. → Times Asked: 2 | Marks: 10
- 6 Assess and justify the concept of Relevance. → Times Asked: 1 | Marks: 10

Module 6 – Applications of Information Retrieval Systems

- 1 Define Multimedia Information Retrieval. Discuss Indexing and Searching. → Times Asked: 3 | Marks: 10 / 20
- 2 Explain Multimedia Indexing Approach. → Times Asked: 2 | Marks: 5 / 10
- 3 Explain Distributed Information Retrieval with architecture. → Times Asked: 3 | Marks: 10
- 4 Discuss Information Retrieval in Digital Libraries. → Times Asked: 2 | Marks: 5 / 10
- 5 Explain Sequential Searching. → Times Asked: 2 | Marks: 5 / 10
- 6 Explain Flat Browsing vs Hypertext Browsing Model. → Times Asked: 3 | Marks: 10 / 20
- 7 Write short notes on Inexact Top K Document Retrieval. → Times Asked: 2 | Marks: 10 / 20